

Homework Submission Policy

- **Late submission:**
 - within 24 hours after its due will incur 20% penalty,
 - after 24 hours and within seven days of its due will incur 50% penalty, and
 - after seven days of its due will not be graded.

Note: One minute late is the same as 23 hours late.

Homework Assignment #1

Textbook Exercises

1.2-17. A poker hand is ...

1.3-13. In the gambling game ...

1.4-15. An urn contains ...

1.5-4. Assume that an insurance company ...

Due: 2:20pm, March 26, 2026

1.2-17. A poker hand is defined as drawing 5 cards at random without replacement from a deck of 52 playing cards. Find the probability of each of the following poker hands:

- (a)** Four of a kind (four cards of equal face value and one card of a different value).
- (b)** Full house (one pair and one triple of cards with equal face value).
- (c)** Three of a kind (three equal face values plus two cards of different values).
- (d)** Two pairs (two pairs of equal face value plus one card of a different value).
- (e)** One pair (one pair of equal face value plus three cards of different values).

1.3-13. In the gambling game “craps,” a pair of dice is rolled and the outcome of the experiment is the sum of the points on the up sides of the six-sided dice. The bettor wins on the first roll if the sum is 7 or 11. The bettor loses on the first roll if the sum is 2, 3, or 12. If the sum is 4, 5, 6, 8, 9, or 10, that number is called the bettor’s “point.” Once the point is established, the rule is as follows: If the bettor rolls a 7 before the point, the bettor loses; but if the point is rolled before a 7, the bettor wins.

- (a) List the 36 outcomes in the sample space for the roll of a pair of dice. Assume that each of them has a probability of $1/36$.
- (b) Find the probability that the bettor wins on the first roll. That is, find the probability of rolling a 7 or 11, $P(7 \text{ or } 11)$.
- (c) Given that 8 is the outcome on the first roll, find the probability that the bettor now rolls the point 8 before rolling a 7 and thus wins. Note that at this stage in the game the only outcomes of interest are 7 and 8. Thus find $P(8 | 7 \text{ or } 8)$.
- (d) The probability that a bettor rolls an 8 on the first roll and then wins is given by $P(8)P(8 | 7 \text{ or } 8)$. Show that this probability is $(5/36)(5/11)$.
- (e) Show that the total probability that a bettor wins in the game of craps is 0.49293. HINT: Note that the bettor can win in one of several mutually exclusive ways: by rolling a 7 or an 11 on the first roll or by establishing one of the points 4, 5, 6, 8, 9, or 10 on the first roll and then obtaining that point on successive rolls before a 7 comes up.

1.4-15. An urn contains 10 red and 10 white balls. The balls are drawn from the urn at random, one at a time. Find the probabilities that the fourth white ball is the fourth, fifth, sixth, or seventh ball drawn if the sampling is done

(a) With replacement.

(b) Without replacement.

(c) In the World Series, the American League (red) and National League (white) teams play until one team wins four games. Do you think that the urn model presented in this exercise could be used to describe the probabilities of a 4-, 5-, 6-, or 7-game series? (Note that either “red” or “white” could win.) If your answer is yes, would you choose sampling with or without replacement in your model? (For your information, the numbers of 4-, 5-, 6-, and 7-game series, up to and including 2012, were 21, 24, 23, 36. This ignores games that ended in a tie, which occurred in 1907, 1912, and 1922. Also, it does not include the 1903 and 1919–1921 series, in which the winner had to take five out of nine games. The World Series was canceled in 1994.)

1.5-4. Assume that an insurance company knows the following probabilities relating to automobile accidents (where the second column refers to the probability that the policyholder has at least one accident during the annual policy period):

Age of Driver	Probability of Accident	Fraction of Company's Insured Drivers
16–25	0.05	0.10
26–50	0.02	0.55
51–65	0.03	0.20
66–90	0.04	0.15

A randomly selected driver from the company's insured drivers has an accident. What is the conditional probability that the driver is in the 16–25 age group?